

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459941.9621 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.056
Transit depth (Rp/Rs)^2	3.62 +/- 2.14 [%]
Orbital Inclination (inc)	86.48 +/- 2.88
Airmass coefficient 1 (a1)	5.79 +/- 3.16
Airmass coefficient 2 (a2)	-1.38 +/- 1.51
Scatter in the residuals of the lightcurve fit is	48.73 %
Best Comparison Star	None
Optimal Aperture	4.06
Optimal Annulus	7.1
Transit Duration (day)	0.0588 +/- 0.0039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.056 vs 0.1594 (deviation 0.42σ)
Duration fit vs published	0.0588 d vs 0.0512 d (deviation 1.51σ)
Mid-time O-C	-4.9 min
Depth SNR	2.6
Residual scatter	48.73 %

Light curve

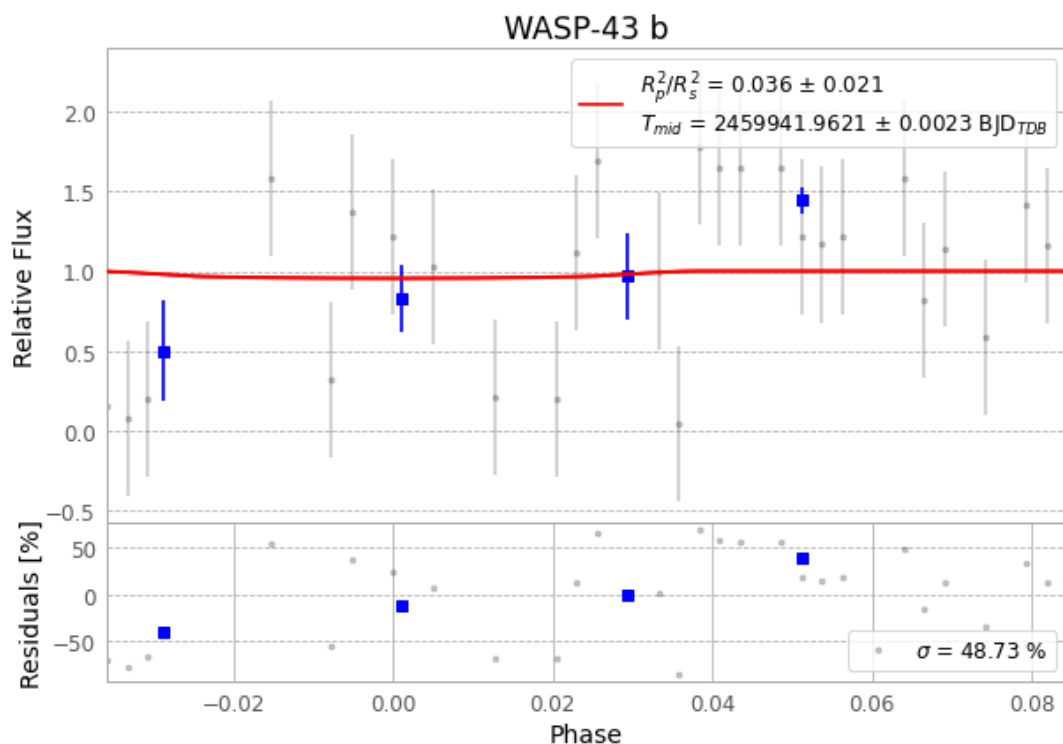


Figure 1 — FinalLightCurve_WASP-43 b_2022-12-28.png (SHA-256 e2360e0ac9e9cef2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [207, 93], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7d893fa5f701ea8b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

51 FITS frames (33 MB), WASP-43221228101215.FITS → WASP-43221228124215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-221228.log (0f5c4fdf0b38...), FinalLightCurve_WASP-43 b_2022-12-28.png (e2360e0ac9e9...), FinalLightCurve_WASP-43 b_2022-12-28.csv (c5bdf1efe1f4...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 22:36Z.